

Introduction to Machine Learning (25737-2)

Problem Set 1

Spring Semester 1402-03

Department of Electrical Engineering

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Due on Esfand 20, 1402 at 23:55



(*) starred problems are optional!

1 Q1

1.1 Normalization constant

Solution: To obtain the normalization constant we need to evaluate the integral

$$\int_{\mathbb{R}^n} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})\right) d\mathbf{x}.$$

Observe that wlog we can assume $\boldsymbol{\mu} = 0$. Thus, it remains to evaluate

$$\int_{\mathbb{R}^n} \exp\left(-\frac{1}{2}\mathbf{x}^T \Sigma^{-1}\mathbf{x}\right) d\mathbf{x} \tag{1}$$

Let $\mathbf{y} = \Sigma^{-1/2}\mathbf{x}$, then by Jacobian transformation (I) can be rewritten as

$$\begin{aligned} \int_{\mathbb{R}^n} \exp\left(-\frac{1}{2}\mathbf{y}^T \mathbf{y}\right) |\Sigma^{1/2}| d\mathbf{y} &= |\Sigma^{1/2}| \prod_{i=1}^n \int_{\mathbb{R}} \exp\left(-\frac{1}{2}y_i^2\right) dy_i \\ &= |\Sigma^{1/2}| \prod_{i=1}^n \sqrt{2\pi} \end{aligned}$$

We use the fact that $\int_{\mathbb{R}} \exp(-\frac{1}{2}x^2) = \sqrt{2\pi}$, then using the fact that $\int p(\mathbf{x}) = 1$, we conclude that the normalization constant is $\frac{1}{\sqrt{(2\pi)^n |\Sigma|}}$.

1.2 Special cases

1.2.1

Solution: By the rule of probability density function for change of variable,

$$\begin{aligned} p_Y(\mathbf{y}) &= \left(\frac{1}{2}\right)^n \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} \exp\left(-\frac{1}{2} \left(\frac{\mathbf{y}}{2} - \mu\right)^T \Sigma^{-1} \left(\frac{\mathbf{y}}{2} - \mu\right)\right) \\ &= \frac{1}{\sqrt{(2\pi)^n |4\Sigma|}} \exp\left(-\frac{1}{2} (\mathbf{y} - 2\mu)^T (4\Sigma)^{-1} (\mathbf{y} - 2\mu)\right) \end{aligned}$$

So Y is still normally distributed, with mean 2μ and covariance matrix 4Σ .

1.2.2

Solution: When Σ is the identity matrix,

$$p_X(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^n}} \exp\left(-\frac{1}{2} \sum_{i=1}^n (x_i - \mu_i)^2\right) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2} (x_i - \mu_i)^2\right)$$

The factorization of the pdf implies that $X_1, \dots, X_n$ are now independent random variables.

1.3 Shape

1.3.1

Solution: Although the covariance matrix is not a (scaled) identity matrix, the off-diagonal elements are still zero, so the two components of the 2D random vector are still independent and the pdf factorizes. Specifically, we have

$$p_X(\mathbf{x}) = \frac{\exp\left(-\frac{1}{2 \times 10} (x_1 - \mu_1)^2\right)}{\sqrt{2\pi \times 10}} \cdot \frac{\exp\left(-\frac{1}{2} (x_2 - \mu_2)^2\right)}{\sqrt{2\pi}}$$

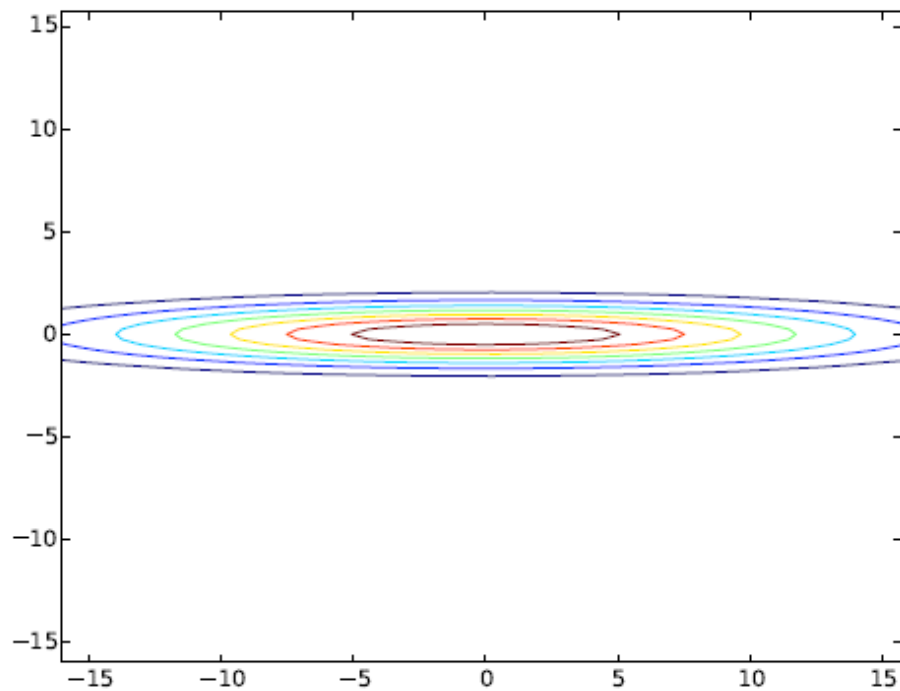
Let

$$c = p_X(\mathbf{x}) = \frac{\exp\left(-\frac{1}{2 \times 10} (x_1 - \mu_1)^2\right)}{\sqrt{2\pi \times 10}} \cdot \frac{\exp\left(-\frac{1}{2} (x_2 - \mu_2)^2\right)}{\sqrt{2\pi}}$$

Via simple manipulation, we get the equation

$$\frac{1}{10}(x_1 - \mu_1)^2 + (x_2 - \mu_2)^2 = C$$

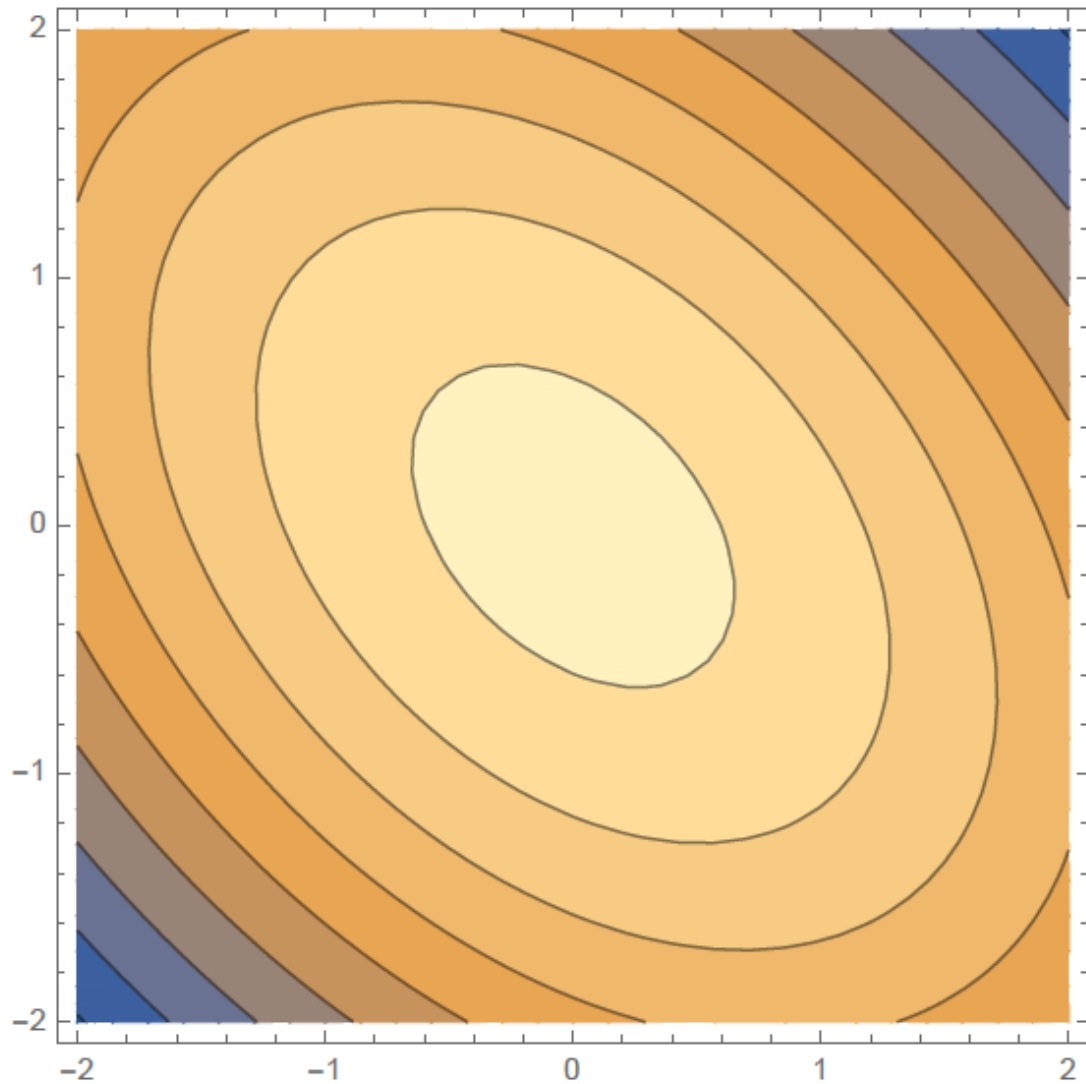
where $C > 0$ is a constant. This is an ellipse in $\mathbb{R}^2$ centered at μ . Specifically, the contours look like those:



1.3.2

Solution: This is a 2D normally distributed random vector. The two components are (negatively) correlated.

Similarly, let $p_X(x) = c$, we will get an equation of an ellipse. The contours for this specific case look like those:



1.3.3

No, because it is not positive semi-definite.

2 types

Determine whether the tasks described below involve supervised learning or unsupervised learning. For supervised learning problems, identify them as regression, classification, or probabilistic classification

2.1

supervised learning, regression

2.2

supervised learning, classification

2.3

supervised learning, probabilistic classification

2.4

unsupervised learning

3

3.1

$$\begin{aligned}
 1.) \quad P(D|\theta) &= \prod_{i=1}^N P(A_i|\theta) = \prod_{i=1}^N \prod_{c=1}^M \theta_c^{a_{ic}} \\
 P(\theta|D) &= \frac{P(D|\theta)P(\theta)}{P(D)} = \frac{1}{P(D)} \prod_{i=1}^N \prod_{c=1}^M \theta_c^{a_{ic}} \frac{1}{\beta(\alpha)} \prod_{j=1}^M \theta_j^{d_j-1} \\
 &\quad A_i \rightarrow \text{one hot encoded vector} \\
 \Rightarrow &= \frac{1}{P(D)} \frac{\prod_{c=1}^M \theta_c^{N_c} \prod_{j=1}^M \theta_j^{d_j-1}}{\beta(\alpha)} = \frac{1}{\beta(\alpha)P(D)} \prod_{c=1}^M \theta_c^{(N_c + d_j - 1)} \sim \text{Dir}(\frac{\alpha+m}{d'}) \\
 m &= \begin{bmatrix} N_1 \\ \vdots \\ N_M \end{bmatrix} \\
 N, x_i - \lambda &= \frac{\sum x_i - N\lambda}{\lambda} \frac{e^{-\lambda}}{c}
 \end{aligned}$$

3.2

$$\begin{aligned}
 2.) \quad P(\lambda|D) &= \frac{P(D|\lambda)P(\lambda)}{P(D)} \quad , \quad P(D|\lambda) = \prod_{i=1}^N \frac{\lambda^{x_i-1} e^{-\lambda}}{x_i!} = \frac{\lambda^{\sum x_i - N\lambda} e^{-N\lambda}}{\prod_{i=1}^N x_i!} \\
 P(\lambda) &= \frac{\beta^\alpha}{\Gamma(d)} \lambda^{d-1} e^{-\beta\lambda} \rightarrow \\
 P(\lambda|D) &= \frac{\lambda^{\sum x_i - N\lambda} e^{-N\lambda}}{\prod_{i=1}^N x_i!} \frac{\beta^\alpha}{\Gamma(d)} \lambda^{d-1} e^{-\beta\lambda} = \frac{\beta^\alpha \lambda^{d-1 + \sum x_i} e^{-(N+\beta)\lambda}}{\prod_{i=1}^N x_i! \Gamma(d)} \\
 \rightarrow P(\lambda|D) &\sim \Gamma(d + \sum x_i, \beta + N) \rightarrow \text{Gamma is CP for Poisson}
 \end{aligned}$$

4

4.1

$$1.) \begin{bmatrix} h_m \\ w_m \end{bmatrix} = \underbrace{\begin{bmatrix} 0.5 & -0.01 \\ 0.001 & 0.3 \end{bmatrix}}_A \begin{bmatrix} h \\ w \end{bmatrix} + \begin{bmatrix} 1 \\ 0.1 \end{bmatrix} + \begin{bmatrix} n_h \\ n_w \end{bmatrix} \rightarrow$$

$$\begin{bmatrix} 2 & 0.067 \\ -0.0067 & 3.33 \end{bmatrix} \begin{bmatrix} 80 \\ 18 \end{bmatrix} = \begin{bmatrix} h \\ w \end{bmatrix} + \begin{bmatrix} 2 & 0.067 \\ -0.0067 & 3.33 \end{bmatrix} \begin{bmatrix} 1 \\ 0.1 \end{bmatrix} + \begin{bmatrix} 2 & 0.067 \\ -0.0067 & 3.33 \end{bmatrix} \begin{bmatrix} n_h \\ n_w \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} h \\ w \end{bmatrix} = \begin{bmatrix} 161.206 - 2.0067 \\ 59.46 - 0.33263 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

$$n_1 = 2n_h + 0.067n_w \rightarrow E(n_1) = 0$$

$$n_2 = -0.0067n_h + 3.33n_w \rightarrow E(n_2) = 0$$

$$\rightarrow \text{Var}(n_1) = E[(2n_h + 0.067n_w)^2] = 4\text{Var}(n_h) + (0.067)^2\text{Var}(n_w)$$

$$\rightarrow \text{Var}(n_1) = 0.08004 \quad \text{Var}(n_2) = (0.0067)^2\text{Var}(n_h) + (3.33)^2\text{Var}(n_w)$$

$$\rightarrow \text{Var}(n_2) = 0.0000009 + 11.1 \times 0.01 = 0.111$$

$$E(n_1 n_2) = -2 \times 0.0067 \text{Var}(n_h) + 3.33 \times 0.067 \text{Var}(n_w) = 6_{12}$$

$$\rightarrow \begin{bmatrix} n_1 \\ n_2 \end{bmatrix} \sim \mu = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \Sigma = \begin{pmatrix} 0.08004 & \delta_{12} \\ \delta_{12} & 0.111 \end{pmatrix}$$

$$\rightarrow \begin{bmatrix} h \\ w \end{bmatrix} \sim \mu + \begin{bmatrix} 159.2 \\ 59.127 \end{bmatrix} = \mu_2 \Sigma$$

in gaussian $\rightarrow$ mode = mean $\rightarrow \boxed{\begin{bmatrix} \hat{h} \\ \hat{w} \end{bmatrix} = \mu_2}$

4.2

2)

$$z \equiv \begin{bmatrix} h \\ w \end{bmatrix} \quad y \equiv \begin{bmatrix} h_m \\ w_m \end{bmatrix} \quad \text{لا توجد علاقة بين الـ LGs}$$

$$y = Wz + b, \quad W = \begin{bmatrix} 0.5 & -0.01 \\ 0.001 & 0.3 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 0.1 \end{bmatrix}$$

$$p(y|z) = N(y|Wz+b, \Sigma_y), \quad \Sigma_y = \begin{bmatrix} 0.02 & 0 \\ 0 & 0.01 \end{bmatrix}$$

$$p(z) = N\left(\begin{bmatrix} 180 \\ 80 \end{bmatrix}, \begin{bmatrix} 8 & 3 \\ 3 & 4 \end{bmatrix}\right)$$

μ_z Σ_z

لا تأخذ في الحسبان العلاقات

$$p(z|y) = N\left(\Sigma_{z|y} (W^T \Sigma_y^{-1} (y-b) + \Sigma_z^{-1} \mu_z), \underbrace{(\Sigma_z^{-1} + W^T \Sigma_y^{-1} W)}_{\Sigma_{z|y}}\right)$$

$$\rightarrow p(z|y) = N\left(\begin{bmatrix} 159.26 \\ 59.62 \end{bmatrix}, \begin{bmatrix} 0.079 & 0.003 \\ 0.003 & 0.107 \end{bmatrix}\right) \rightarrow$$

$$\begin{bmatrix} \hat{h} \\ \hat{w} \end{bmatrix} = \begin{bmatrix} 159.26 \\ 59.62 \end{bmatrix} \rightarrow \text{closer to prior than part 1}$$

4.3

3.) here the noise is zero mean and zero var $\rightarrow n=0 \rightarrow$
the problem is deterministic

$$\rightarrow \begin{bmatrix} h_m \\ w_m \end{bmatrix} = \begin{bmatrix} 0.5 & -0.01 \\ 0.001 & 0.3 \end{bmatrix} \begin{bmatrix} h \\ w \end{bmatrix} + \begin{bmatrix} 1 \\ 0.1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} h_m \\ w_m \end{bmatrix} = \begin{bmatrix} 80 \\ 18 \end{bmatrix} \rightarrow$$

$$\begin{bmatrix} h \\ w \end{bmatrix} \sim \begin{cases} f_h(x) = \delta(x - 159.2) \\ f_w(x) = \delta(x - 59.127) \end{cases} \rightarrow \begin{matrix} \hat{h} = 159.2 \\ \hat{w} = 59.127 \end{matrix}$$

4.4

$$4.) \quad \Sigma_y = \begin{bmatrix} \sigma^2 & 0 \\ 0 & \sigma^2 \end{bmatrix} = \sigma^2 I \rightarrow \Sigma_y^{-1} = \frac{1}{\sigma^2} I \quad \rightarrow \Sigma_{z|y} = \left(\Sigma_z^{-1} + \frac{1}{\sigma^2} W^T W \right)^{-1}$$

function of σ^2

$$\rightarrow p(z|y) = N \left(\underbrace{\Sigma_{z|y} \left(\frac{1}{\sigma^2} W^T (y-b) + \Sigma_z^{-1} \mu_z \right)}_{\mu_{z|y}}, \Sigma_{z|y} \right)$$

$$\rightarrow \begin{bmatrix} \hat{h} \\ \hat{m} \end{bmatrix} = \mu_{z|y} = \begin{bmatrix} f(\sigma^2) \\ g(\sigma^2) \end{bmatrix}$$

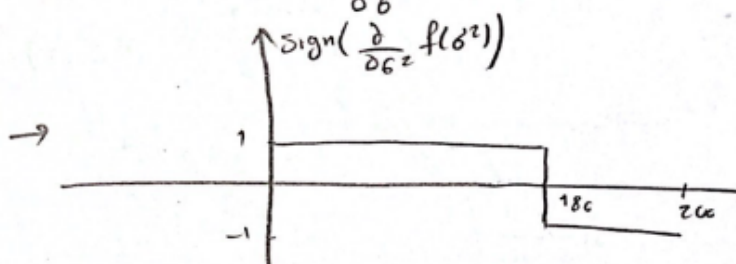
It is also obvious that if we measure our highland weight multiple times with the same prior the only thing that affects the estimation is measurement noise.

$$b.) \quad \text{if } \hat{h} > 180 \rightarrow \frac{\partial}{\partial \sigma^2} f(\sigma^2) < 0 \rightarrow \sigma^2 \uparrow \rightarrow \hat{h} \downarrow$$

prior mean

$$\text{if } \hat{h} < 180 \rightarrow \frac{\partial}{\partial \sigma^2} f(\sigma^2) > 0 \rightarrow \sigma^2 \uparrow \rightarrow \hat{h} \uparrow$$

$$\rightarrow [0, 180) \rightarrow \frac{\partial}{\partial \sigma^2} f(\sigma^2) > 0, \quad [180, 200] \rightarrow \frac{\partial}{\partial \sigma^2} f(\sigma^2) < 0$$



assume $y-b$ and all other factors in $p(z|y)$ constant and calculate $f(\sigma^2)$ you can see the above result.

5

You are responsible for encrypting Nazi messages during World War II. The number of messages you are given to encrypt each day can be modeled by a random variable ($N \sim \text{Poi}(\lambda)$). Your device is a bit old, and it encrypts each message independently with a probability of error p . The number of correctly encrypted messages in a day is represented by the random variable X and the number of erroneous messages in a day is represented by the random variable Y .

5.1

Find the probability mass function of X .

$$\begin{aligned}
P(X = i) &= \sum_{n=i}^{\infty} P(X = i|N = n)P(N = n) = \sum_{n=i}^{\infty} \binom{n}{i} p^i (1-p)^{n-i} \frac{e^{-\lambda} \lambda^n}{n!} \\
P(X = i) &= \frac{(\lambda p)^i e^{-\lambda}}{i!} \sum_{j=0}^{\infty} \frac{((1-p)\lambda)^j}{j!} = \frac{(\lambda p)^i e^{-\lambda}}{i!} (e^{(1-p)\lambda}) \\
&\longrightarrow X \sim Poi(\lambda p)
\end{aligned}$$

5.2

Find the joint probability mass function of the two random variables X and Y .

Assume that $Y + X = N$:

$$\begin{aligned}
P(X = i, Y = j) &= \sum_{n=i+j}^{\infty} P(X = i, Y = j|N = n)P(N = n) \\
&= P(X = i, Y = j|N = i+j)P(N = i+j) \\
X = i, N = i+j &\rightarrow Y = j, q = 1-p \Rightarrow P(X = i, Y = j|N = i+j) = P(X = i|N = i+j) \\
P(X = i, Y = j) &= P(X = i|N = i+j)P(N = i+j) = \binom{i+j}{i} p^i q^j \frac{e^{-\lambda(p+q)} \lambda^{(i+j)}}{(i+j)!} \\
&= \frac{(i+j)!}{i!j!} p^i q^j \frac{e^{-\lambda(p+q)} \lambda^{(i+j)}}{(i+j)!} = \frac{e^{-\lambda p} (\lambda p)^i}{i!} \frac{e^{-\lambda q} (\lambda q)^j}{j!} \\
&\longrightarrow P(X = i, Y = j) = \frac{e^{-\lambda p} (\lambda p)^i}{i!} \frac{e^{-\lambda q} (\lambda q)^j}{j!} \\
&\implies Y \sim Poi(\lambda q)
\end{aligned}$$

5.3

Are X and Y independent?

Since $P(X = i, Y = j) = P(X = i)P(Y = j)$ we conclude that X and Y are independent.

6

6.1

Solution:

Let X be a random variable. To prove this inequality, we use ****Markov's inequality****. Markov's inequality states that for any non-negative function $f(X)$:

$$P(X \geq t) = P(f(X) \geq f(t)) \leq \frac{\mathbb{E}[f(X)]}{f(t)}.$$

Here, we choose the function $f(X) = e^{sX}$.

By Markov's inequality, we get:

$$P(X > t) = P(e^{sX} > e^{st}) \leq \frac{\mathbb{E}[e^{sX}]}{e^{st}}.$$

This is exactly the Chernoff inequality:

$$P(X > t) \leq e^{-st} \mathbb{E}[e^{sX}].$$

Thus, Chernoff's inequality is proven.

6.2

First show that for $X \sim \mathcal{N}(0, \sigma^2)$ and for each $s \in \mathbb{R}$, we have:

$$\mathbb{E}[e^{sX}] \leq e^{\frac{s^2\sigma^2}{2}}.$$

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{x^2}{2\sigma^2}}.$$

The expectation becomes:

$$\mathbb{E}[e^{sX}] = \int_{-\infty}^{\infty} e^{sx} f_X(x) dx = \int_{-\infty}^{\infty} e^{sx} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{x^2}{2\sigma^2}} dx.$$

This simplifies to:

$$\frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{\frac{2sx\sigma^2 - x^2}{2\sigma^2}} dx = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{(x-s\sigma^2)^2}{2\sigma^2} + \frac{s^2\sigma^2}{2}} dx.$$

We can factor out the exponential term:

$$= \frac{e^{\frac{s^2\sigma^2}{2}}}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{(x-s\sigma^2)^2}{2\sigma^2}} dx.$$

This integral evaluates to 1, so we get:

$$\mathbb{E}[e^{sX}] \leq e^{\frac{s^2\sigma^2}{2}}.$$

Show that if X is a random variable that has the property from the previous part with parameter σ^2 (note that X is not necessarily Gaussian), then for each $t > 0$, we have:

$$\mathbb{P}(|X| \geq t) \leq 2 \exp\left(-\frac{t^2}{2\sigma^2}\right).$$

Solution: From Chebyshev's inequality, we know that:

$$\mathbb{P}(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

It follows that for the random variable X with the aforementioned properties, we have:

$$\mathbb{P}(|X| \geq t) \leq \frac{\sigma^2}{t^2}.$$

Thus,

$$\mathbb{P}(|X| \geq t) = \mathbb{P}(X \geq t) + \mathbb{P}(X \leq -t) \leq \mathbb{P}(X \geq t) + \mathbb{P}(-X \geq t).$$

Next, by using Markov's inequality, we have:

$$\mathbb{P}(e^{sX} \geq e^{st}) + \mathbb{P}(e^{-sX} \geq e^{st}) \leq \frac{\mathbb{E}[e^{sX}]}{e^{st}} + \frac{\mathbb{E}[e^{-sX}]}{e^{st}} = e^{-st} (\mathbb{E}[e^{sX}] + \mathbb{E}[e^{-sX}]).$$

Using the assumption from the previous part, we can obtain:

$$\mathbb{P}(|X| \geq t) \leq 2e^{-st + \frac{s^2\sigma^2}{2}}.$$

By setting $s = \frac{t}{\sigma^2}$, we get:

$$\mathbb{P}(|X| \geq t) \leq 2e^{-\frac{t^2}{2\sigma^2}},$$

thus proving the claim.

6.3

Show that for any random variable Z , we have:

$$\mathbb{E}[\max(0, Z)] = \int_0^\infty \mathbb{P}[Z \geq x] dx.$$

Solution: We can write:

$$\mathbb{E}[\max(0, Z)] = \int_{-\infty}^\infty \max(0, Z) f_Z(z) dz.$$

Now, notice that:

$$\max(0, Z) = \begin{cases} Z & \text{if } z \geq 0, \\ 0 & \text{if } z < 0. \end{cases}$$

Thus, we can simplify the integral to start from 0:

$$\mathbb{E}[\max(0, Z)] = \int_0^\infty z f_Z(z) dz.$$

Finally, notice that for each fixed z , we can rewrite it as $z = \int_0^z dx$. A change of the order of integration can prove the inequality:

$$\mathbb{E}[\max(0, Z)] = \int_0^\infty z f_Z(z) dz = \int_0^\infty \int_0^x dx f_Z(z) dz = \int_0^\infty \int_z^\infty f_Z(x) dx dz = \int_0^\infty \mathbb{P}[Z \geq z] dz.$$

6.4

$$P_{X|\varepsilon}(x) = \mathbf{1}\{x \in \varepsilon\} \frac{P_X(x)}{\delta} \implies \mathbb{E}[f(X)|\varepsilon] = \frac{1}{\delta} \mathbb{E}[f(X) \times \mathbf{1}\{X \in \varepsilon\}]$$

Therefore, we have:

$$\mathbb{E}[f(X)|\varepsilon] = \frac{1}{\delta} \mathbb{E}[f(X) \times \mathbf{1}\{X \in \varepsilon\}] = \frac{1}{\delta} \mathbb{E}[(f(X) - \mathbb{E}[f(X)] + \mathbb{E}[f(X)]) \times \mathbf{1}\{X \in \varepsilon\}]$$

As a result:

$$\mathbb{E}[f(X)|\varepsilon] = \frac{1}{\delta} \mathbb{E}[(f(X) - \mathbb{E}[f(X)]) \times \mathbf{1}\{X \in \varepsilon\}] + \mathbb{E}[f(X)]$$

By the Cauchy-Schwarz inequality, we get:

$$\implies \mathbb{E}[f(X)|\varepsilon] \leq \mathbb{E}[f(X)] + \frac{1}{\delta} \times \sqrt{\delta \operatorname{Var}(f(X))}$$

Thus:

$$\mathbb{E}[f(X)|\varepsilon] \leq \mathbb{E}[f(X)] + \sqrt{\frac{\operatorname{Var}(f(X))}{\delta}}$$

7

Given the way Y is produced, we know that $Y \in \{X, X+1, \dots, 6\}$. Moreover, the die is fair. Therefore, given X , Y is uniformly distributed among the possible outcomes:

7.1

$$\begin{aligned} X = 2 &\implies Y \in \{2, 3, 4, 5, 6\} \\ &\implies \boxed{\mathbb{P}(Y = 6|X = 2) = \frac{1}{5}} \end{aligned}$$

7.2

By the Partition Rule, we have:

$$\begin{aligned}\mathbb{P}(Y = 6) &= \sum_{i=1}^6 \mathbb{P}(Y = 6|X = i)\mathbb{P}(X = i) \\ &= \sum_{i=1}^6 \frac{1}{7-i} \times \frac{1}{6} = \frac{1}{6} \times \left(\frac{1}{1} + \frac{1}{2} + \cdots + \frac{1}{6} \right) = \boxed{\frac{49}{120}}\end{aligned}$$

7.3

By Bayes' Theorem, we have:

$$\begin{aligned}\mathbb{P}(X = 2|Y = 6) &= \frac{\mathbb{P}(Y = 6|X = 2)\mathbb{P}(X = 2)}{\sum_{i=1}^6 \mathbb{P}(Y = 6|X = i)\mathbb{P}(X = i)} \\ \implies \mathbb{P}(X = 2|Y = 6) &= \frac{\frac{1}{5} \times \frac{1}{6}}{\frac{49}{120}} = \frac{4}{49}\end{aligned}$$

8

8.1

- **Sensitivity**, or the sensitivity of the diagnostic test, is the probability that the test will be positive when the person is actually diseased. In other words, if a person is diseased, the probability that the diagnostic test will be positive is equal to **Sensitivity**. In this case, **Sensitivity** is 0.875.
- **Specificity**, or the specificity of the diagnostic test, is the probability that the test will be negative when the person is not diseased. In other words, if a person is not diseased, the probability that the diagnostic test will be negative is equal to **Specificity**. In this case, **Specificity** is 0.975.

Therefore, the **Confusion Matrix** is as follows:

	Positive	Negative
Positive	True Positive	False Negative
Negative	False Positive	True Negative

Which in this case is as follows:

	Positive	Negative
Positive	0.875	0.125
Negative	0.025	0.975

8.2

Considering that we know the diagnostic test has specific **Sensitivity** and **Specificity**, we can use Bayes' theorem to calculate the probability that a person is diseased.

The probability of a person being diseased in the population is 0.1. Also, the probability that the test will be positive when the person is diseased is equal to **Sensitivity**, and the probability that the test will be negative when the person is not diseased is equal to **Specificity**.

Therefore, the probability that the test will be positive when the person is diseased is 0.875, and the probability that the test will be negative when the person is not diseased is 0.975.

Thus, the probability that a person is diseased given that the test result is as follows is:

Positive Test:

$$\begin{aligned} P(\text{Covid-19}|\text{Positive Test}) &= \frac{P(\text{Positive Test}|\text{Covid-19})P(\text{Covid-19})}{P(\text{Positive Test})} \\ &= \frac{0.875 \times 0.1}{0.875 \times 0.1 + 0.025 \times 0.9} \simeq 0.795 \end{aligned}$$

Negative Test:

$$\begin{aligned} P(\text{Covid-19}|\text{Negative Test}) &= \frac{P(\text{Negative Test}|\text{Covid-19})P(\text{Covid-19})}{P(\text{Negative Test})} \\ &= \frac{0.125 \times 0.1}{0.125 \times 0.1 + 0.975 \times 0.9} \simeq 0.014 \end{aligned}$$

Now, assuming that a person with the disease will definitely die, we have:

$$\begin{aligned} \text{cost-noop} &= P(\text{Covid-19}) \times \text{Years expected to live} \\ \text{cost-treatment} &= 8 \end{aligned}$$

Where **cost-noop** represents the cost of not treating, and **cost-treatment** represents the cost of treatment.

Finally, the following table can be examined for the individuals in question:

Age	Test Result	Probability of Covid-19	cost-noop	cost-treatment
20	Negative	0.014	$60 \times 0.014 = 0.84$	8
20	Positive	0.795	$60 \times 0.795 = 47.7$	8
70	Negative	0.014	$10 \times 0.014 = 0.14$	8
70	Positive	0.795	$10 \times 0.795 = 7.95$	8

As can be seen from the results in the table, only for 20-year-olds who have tested positive is prescribing treatment medication logical.

8.3

Based on the results from the previous section, we have:

$$\begin{aligned} P(\text{Covid-19}|\text{Positive Test}) \times \text{Years expected to live} &\geq \text{cost-treatment} \\ \implies 0.795 \times YL &\geq 8 \implies YL \geq 10.06 \end{aligned}$$

Therefore, the age threshold for prescribing medication is $70 - 10.06 = 59.94$ years.

8.4

Based on the results from the previous sections, it is clear that the new medication should be prescribed for a 60-year-old individual who tested positive and for a 20-year-old under any test result.

9

9.1

This occurs because of how the data is distributed, an example of Simpson's Paradox. Treatment A was disproportionately used on patients with large stones (263/350), which are harder to treat and have lower success rates overall. Treatment B was disproportionately used on patients with small stones (270/350), which are easier to treat and have higher success rates overall. When aggregating, the higher volume of easier cases for Treatment B artificially inflates its overall success rate compared to Treatment A, despite Treatment A being more effective within each stone size category.

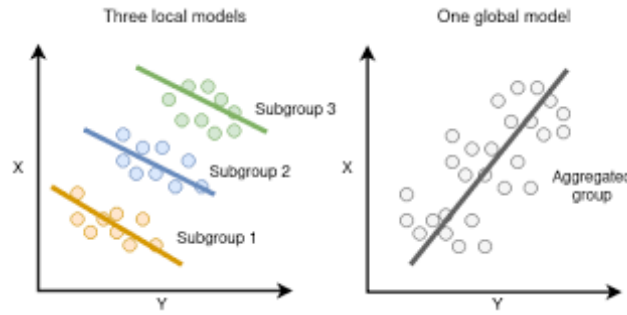


Figure 1: A continuous example

9.2

The confounding factor is the stone size. Stone size is associated with both the choice of treatment (Treatment A for large, Treatment B for small) and the outcome (success rates are lower for large stones than for small stones). Because the distribution of this confounding factor (stone size) differs significantly between the two treatment groups, the aggregated overall results are misleading.

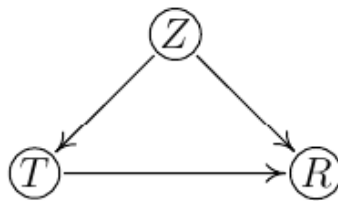


Figure 2: Relationships between Stone Size, Treatment, and Recovery. Arrows indicate influence. We see that Stone Size (Z) impacts the Treatment decision (T) (e.g., larger stones get Treatment A) and also independently affects the chance of Recovery (R) (larger stones are harder to treat). The Treatment (T) itself also impacts Recovery (R). This structure helps explain the surprising overall statistics.

While Treatment B is more correlated with 'R' in the aggregate data, this association is misleading, and actively implementing Treatment B based on it would be ineffective/suboptimal.

9.3

Advise them to ignore the overall success rates as they are misleading due to the confounding effect of stone size and the non-random assignment of treatments. Decisions should be based on the subgroup analysis. Since Treatment A shows a higher success rate for both small stones (93% vs 87%) and large stones (73% vs 69%), it is the more effective treatment regardless of stone size, based purely on success rate. Therefore, for a given patient, Treatment A should be recommended over Treatment B, assuming other factors are equal.